Course Title	Statistical Methods (Lab:python)		
Course Code	DS602	Credit Structure	3-0-2-4
Course Format	Three 50 minutes lecture and 2 hours of Lab each week		
Category	Core	Semester	Autumn Semester (AY 2026-27)
Program	UG/PG		
Prerequisites (If any)			
Course Objectives/ Brief Course Description	This course provides a rigorous foundation in statistical theory, probability modeling, and computational techniques for data analysis. Students will learn to summarize and visualize data, apply standard probability models, and use Monte Carlo simulations to understand sampling distributions. The curriculum equips students to perform statistical inference via parameter estimation and hypothesis testing, build predictive models using linear and logistic regression, and analyze time series data. By integrating theoretical concepts with modern computational methods like Bootstrap resampling, students will develop the essential skills to model uncertainty and draw robust, actionable conclusions from complex datasets.		
Evaluation/ Grading Policy	• Lab/Assignments: 20% • In-Semester exams I and II : 20% +20% • End semester examination: 40% • Relative grading		
Course Materials/ References	• Probability and Statistics for Engineers and Scientists by Sheldon Ross • An introduction to probability theory and its application (Volume -1) — William Feller • Statistical Inference – George Casella and Roger L. Berger • An introduction to the Bootstrap—Bradley Efron and Robert J. Tibshirani		
Detailed Course Content	APPENDIX – I		

Course Outcome:

CO1: Learn the fundamental concepts and the basic tools of Data Analytics

CO2: Learn the implementation of the analytics tools using Python

CO3: Acquire the skill to analyze data in a systematic and logical manner to answer important research questions arising in different disciplines.

Program Outcomes (Mark it X if applicable refer Appendix – II)

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X		X	X							X

APPENDIX- I: Detailed Course Content (Session-wise/ Module-wise)

Module	Title & Description	No of Lectures
1.	**Data summarization** : Data types, Data visualization and summarization with the help of tables and the important descriptive measures.	**5**
2.	**Probability :** Concepts of probability (both objective and subjective). Statistical interpretation of probability, Bayes' theorem and its applications in solving real-life decision problems	**5**
3.	**Standard probability distributions :** Standard probability models (Binomial, Poisson, Uniform, Exponential, Gaussian), Probability distributions of transforms of random variables, Data generation from probability distributions using Monte Carlo method.	**6**
4.	**Sampling :** Concepts of population and sample (both random and nonrandom), Parameter & Statistic, Probability distribution of a statistic (small sample & large sample), Central limit theorems, Illustration of sampling distribution using Monte Carlo simulation.	**5**
5.	**Statistical estimation:** Point and interval estimation, Methods of estimation (method of moments & maximum likelihood method)	**3**
6.	**Statistical testing of point null hypotheses:** Null and alternative hypotheses, Significance testing using p-value, Some standard parametric and non-parametric tests	**4**
7.	**Regression Analysis:** Linear Regression & Logistic Regression, Analysis of Variance	**6**

8.	**Time Series Analysis:** Components of time series, Determination of Trend and Seasonal components Special topics	**3**
9.	**Bootstrap estimation of Bias and Standard Error**	**2**

Appendix – II : Program Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need

for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.